

Data Storage

Everything is stored in a single SQL Server database via Entity Framework Core. All tables follow two conventions automatically applied by `ApplicationDbContext`:

- **Audit fields:** `CreatedAt`, `UpdatedAt`, `CreatedBy`, `UpdatedBy` are stamped on every save.
 - **Soft delete:** deletions set `IsDeleted = true` and `DeletedAt` rather than removing the row.
-

Users

Table: `Users` (ASP.NET Core Identity `IdentityUser<Guid>` extended)

Field	Type	Notes
Id	Guid	PK
Email	string	Unique, used as username
PhoneNumber	string?	Unique across all users
FullName	string	Display name
Language	string	"en" or "ar"
ProfileImage	string?	URL

Status	enum	Active / Suspended / Banned / PendingVerification
OrderUpdatesEnabled	bool	Notification preference
MarketingNotificationsEnabled	bool	Notification preference
CreatedAt	DateTimeOffset	
UpdatedAt	DateTimeOffset?	
PasswordHash	string	Managed by Identity
SecurityStamp	string	Managed by Identity

Supporting Identity tables: Roles, UserRoles, UserClaims, UserLogins, UserTokens, RoleClaims.

Roles seeded at startup: Customer, Merchant, Charity, Admin.

Refresh Tokens

Table: RefreshTokens

Field	Type	Notes
Id	Guid	PK
UserId	Guid	FK → Users
Token	string (500)	64-byte random, Base64
ExpiresAt	DateTimeOffset	Now + 30 days

RevokedAt	DateTimeOffset?	Set on logout, rotation, or reuse detection
ReplacedByToken	string?	The new token string that superseded this one
CreatedByIp	string?	Client IP at issuance
RevokedByIp	string?	Client IP at revocation

A token is **active** when `RevokedAt == null AND ExpiresAt > now`. Each use rotates the token (old is revoked, new is issued).

Stores

Table: `Stores`

Field	Type	Notes
Id	Guid	PK
OwnerId	Guid	FK → Users
Name	string (150)	English name
NameAr	string? (150)	Arabic name
Description	string?	English description
DescriptionAr	string?	Arabic description
Logo	string?	URL
Phone	string?	Contact phone
Email	string?	Contact email
BusinessCategory	enum?	Restaurant / Bakery / Supermarket / etc.

Governorate	string? (100)	Set in onboarding step 2
City	string? (100)	
Neighborhood	string? (100)	
Street	string? (200)	
BuildingNo	string?	Building / landmarks
Latitude	double?	
Longitude	double?	
OpeningHours	string?	JSON-encoded weekly schedule
VerificationStatus	enum	Unverified → Pending → Verified / Rejected
AverageRating	double	Computed from reviews
IsDeleted	bool	Soft delete

A draft store is created at registration for Merchant and Charity accounts. The store moves through verification states as documents are uploaded and reviewed.

Store Verification Documents

Table: `StoreVerifications`

Three document slots per store, matching the `document_upload_step_2` screen:

Document type	Meaning
---------------	---------

CommercialRegistration	Official business registration certificate
TaxIdCertificate	Tax identification document
StoreFacilityPhoto	Exterior photograph of the store

Field	Type	Notes
Id	Guid	PK
StoreId	Guid	FK → Stores
VerificationType	string (100)	One of the three types above
DocumentUrl	string (500)	Path served as static file
Status	enum	Pending / Verified / Rejected
ReviewedBy	Guid?	FK → Users (admin who reviewed)
ReviewedAt	DateTimeOffset?	

Once all three document types are uploaded, `Store.VerificationStatus` moves to `Pending` automatically.

Product Listings

Table: `ProductListings`

Field	Type	Notes
Id	Guid	PK
StoreId	Guid	FK → Stores
CategoryId	Guid	FK → Categories
Title	string (200)	English title

TitleAr	string? (200)	Arabic title
Description	string?	English description
DescriptionAr	string?	Arabic description
OriginalPrice	decimal (10,2)	Price before discount
DiscountedPrice	decimal (10,2)	AI-recommended or merchant-set price
QuantityAvailable	int	Current stock
ExpirationDate	DateOnly	The core data point for the platform — drives discount urgency
Status	enum	Active / Inactive / SoldOut / Expired
IsDeleted	bool	Soft delete

The `ExpirationDate` is how the platform works — items close to expiry get deeper discounts. The AI reads this field to calculate discount recommendations.

Product Images

Table: `ProductImages`

Field	Type	Notes
Id	Guid	PK
ListingId	Guid	FK → ProductListings
ImageUrl	string (500)	
DisplayOrder	int	Ordering for carousel

AI Recognition Results

Table: AIRecognitionResults

Stores the output of OCR / GPT Vision analysis per product listing.

Field	Type	Notes
Id	Guid	PK
ListingId	Guid	FK → ProductListings (1-to-1)
DetectedProduct	string? (200)	Product name extracted from image
ExtractedText	string? (4000)	All text found on the label
ExtractedExpiryDate	DateOnly?	Expiry date read from the label
ConfidenceScore	double	0.0 – 1.0
Reviewed	bool	Whether a human has confirmed this result

This table is what bridges the physical product label to the digital listing. The `ConfidenceScore` drives whether the AI acts autonomously or queues the result for human review.

Categories

Table: Categories

Field	Type	Notes
Id	Guid	PK
Name	string (100)	English, unique

NameAr	string? (100)	Arabic
Icon	string?	Icon identifier

Orders

Table: `Orders`

Field	Type	Notes
Id	Guid	PK
UserId	Guid	FK → Users (buyer)
TotalAmount	decimal (10,2)	
PaymentStatus	enum	Pending / Paid / Failed / Refunded
OrderStatus	enum	Pending / Confirmed / Ready / PickedUp / Cancelled

Table: `OrderItems`

Field	Type	Notes
OrderId	Guid	PK (composite)
ListingId	Guid	PK (composite)
Quantity	int	
UnitPrice	decimal (10,2)	Price at time of purchase

Payments

Table: `Payments` (1-to-1 with Orders)

Field	Type	Notes
Id	Guid	PK
OrderId	Guid	FK → Orders (unique)
Amount	decimal (10,2)	
Method	string (50)	e.g. "Card", "Cash", "Wallet"
Status	enum	Pending / Completed / Failed / Refunded
TransactionReference	string? (200)	Payment gateway reference

Reviews

Table: `Reviews`

Field	Type	Notes
Id	Guid	PK
StoreId	Guid	FK → Stores
UserId	Guid	FK → Users
OrderId	Guid	FK → Orders (unique — one review per order)
Rating	int	1–5
Comment	string? (1000)	

Addresses

Table: Addresses

Field	Type	Notes
Id	Guid	PK
UserId	Guid	FK → Users
AddressType	enum	Home / Company
City	string (100)	
District	string (100)	
Street	string (200)	
BuildingNo	string? (20)	
Floor	string? (20)	
ApartmentNo	string? (20)	
Notes	string? (300)	Landmarks / special instructions
Latitude	double	
Longitude	double	
IsDefault	bool	One default address per user

Notifications

Table: Notifications

Field	Type	Notes
Id	Guid	PK
UserId	Guid	FK → Users
Title	string (200)	

Body	string (1000)	
Type	string (50)	e.g. "OrderUpdate", "Promotion", "VerificationResult"
IsRead	bool	

Support Tickets

Table: SupportTickets

Field	Type	Notes
Id	Guid	PK
UserId	Guid	FK → Users
Category	string	e.g. "Delivery", "Payment", "Account"
Priority	enum	Low / Normal / High / Urgent
Status	enum	Open / InProgress / Resolved / Closed

Table: TicketMessages

Field	Type	Notes
Id	Guid	PK
TicketId	Guid	FK → SupportTickets
Body	string	Message content
SentBy	Guid	User or admin ID

What the AI stores (summary)

The critical AI-stored data lives in `AIRecognitionResults`:

- **ExtractedExpiryDate** — read from the product label image by OCR/GPT Vision. This is the most important field — it's what the pricing algorithm uses to calculate how urgent the discount is.
- **DetectedProduct** — product name extracted from the label.
- **ExtractedText** — full raw text from the label, kept for auditing and re-processing.
- **ConfidenceScore** — determines whether the AI's output is trusted autonomously or sent to a human moderator.

The discount recommendation itself lives in

`ProductListing.DiscountedPrice`. The AI writes this field when it has sufficient confidence; otherwise it proposes a value and waits for merchant confirmation.